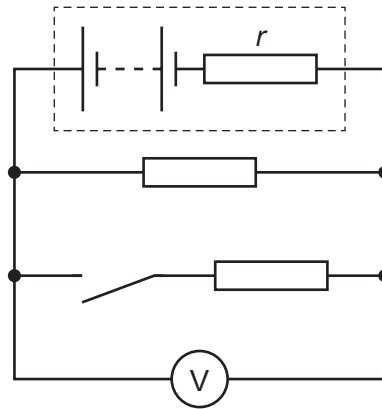


- 37 The circuit diagram shows a battery with internal resistance r , two resistors, a switch and a voltmeter. The switch is open.



The switch is now closed.

What happens to the current in the battery, and what happens to the reading on the voltmeter, when the switch is closed?

	current in battery	reading on voltmeter
A	increases	decreases
B	increases	remains the same
C	remains the same	decreases
D	remains the same	remains the same

- 38 What is the correct equation for β^+ decay?

- A** neutron $\rightarrow$ proton + electron + electron antineutrino
B neutron $\rightarrow$ proton + electron + electron neutrino
C proton $\rightarrow$ neutron + positron + electron antineutrino
D proton $\rightarrow$ neutron + positron + electron neutrino

- 39 A nucleus of polonium, $^{214}_{84}\text{Po}$, decays by emitting an α -particle to become a nucleus of lead.

The nucleus of lead is also unstable and decays by emitting a β^- particle to form a nucleus of bismuth which then decays by emitting a β^- particle to produce a nucleus X.

What is the number of neutrons in nucleus X?

- A** 126 **B** 128 **C** 130 **D** 210